

Diagrammatic Monte Carlo Computational Lab

Interactions and Correlations in Condensed Matter

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1. **Where:** AULA INFORMATICA ZOOLOGIA, second floor, via F. Selmi 3. The PC Lab Room is equipped with dual-booting PC: linux and window (linux recommended). Login to Lab PC.

Linux: Select **ubuntu** at the beginning of the booting (first line)

Username: STUDENTI^name.surname00

Windows: default choice at booting time

Username: your complete unibo email address

2. **Connection to the cluster:** Simulations will be physically performed in the OPH cluster. The cluster is accessible from the PCs using SSH.

Type in the terminal window (in LINUX or WINDOWS) the following command:

```
ssh -J name.surname00@bastion-nav.difa.unibo.it name.surname00@ophfe1
```

name.surname00 = your unibo studio account (e.g. cesare.franchini2)

password: your default Unibo password

mind: the password is required twice

This will do a two-step connection, first to bastion-nav.difa.unibo.it, then to ophfe1.

To see images and plots from the remote server, avoid disconnections see further details about **ssh** at the end of the guide (section 7).

3. **Setup working directory:** Each student should copy the lab directory in their own home directory:

```
cp -r /home/STUDENTI/marco.barducci2/DIAGMC-LAB .
```

The directory contains:

- **attractive_potential** : attractive potential hamiltonian
- **blade_run** : script to run the job
- **env_setup** : setup environment to run the job
- **free_propagator** : free propagator hamiltonian
- **froehlich** : Froehlich polaron hamiltonian
- **two_level_system** : two level system hamiltonian

The codes use python for post processing To create a python environment with the necessary packages for data analysis type:

```
source ~/DIAGMC-LAB/env_setup
```

Content of `env_setup` file:

```
# python environment for post-processing

env=~/.DIAGMC-PY
python3 -m venv $env
source $env/bin/activate
pip install --upgrade pip
pip install numpy matplotlib scipy
```

This will create a python environment called `.DIAGMC-PY` and install the packages to be used later.

4. How to run and control a job

To start any job you should use the script `blade_run` located in `~/DIAGMC-LAB/blade_run`.

To run a Monte Carlo simulation first enter your `DIAGMC-LAB` folder
(from your home i.e. `/home/STUDENTI/name.surname00`)

```
cd DIAGMC-LAB
```

Here you can edit the jobscript `blade_run` (see section 6).

You can use an editor such as `nano` or `vim`.

Important commands:

```
- sbatch: submit job (i.e. sbatch blade_run)
- squeue: check status (squeue -u username)
- scancel: cancel a job (scancel #jobnumber)
```

More info can be found on the wiki <https://apps.difa.unibo.it/wiki/oph:cluster:jobs>

5. DIAGMC-LAB folder

Each exercise folder contains an executable with the same and a `test` subfolder where the exercise will be done. Additional python scripts for post-processing are already inside `test`.

```
DIAGMC-LAB/
|-- attractive_potential
|   |-- attractive_potential
|   '-- test
|       '-- plot_fit.py
|-- blade_run
|-- env_setup
|-- free_propagator
|   |-- free_propagator
|   '-- test
|       '-- plot_histogram.py
|-- froehlich
|   |-- froehlich
|   '-- test
|       '-- post_processing.py
'-- two_level_system
    |-- test
    |-- post_processing.py
    '-- two_level_system
```

At the root there are two files: `blade_run` which is the script used to submit calculations (`sbatch blade_run`) and the `env_setup` script already discussed.

6. Setup a job

The job execution is controlled via the `blade_run` script.

```
1  #!/bin/bash
2  #SBATCH --constraint=blade      ## only blade nodes
3  #SBATCH --nodes=1
4  #SBATCH --tasks-per-node=2
5  #SBATCH --mem-per-cpu=1G
6  #SBATCH --time=00:05:00
7  #SBATCH --exclude=bld15,bld16
8  #SBATCH --output=a_%N-%j.out
9
10 source /home/software/materia/intel/oneapi/setvars.sh --force
11 source ~/.DIAGMC-PY/bin/activate
12 export LD_LIBRARY_PATH=/home/STUDENTI/marco.barducci2/apps/gcc-11.4.0/
13   lib64:${LD_LIBRARY_PATH}
14 export UCX_TLS=ud,sm,self
15
16 task=free_propagator           # modify for another exercise (e.g. froehlich)
17 pp=plot_histogram.py           # modify when different (e.g. post_processing.py)
18
19 cd $task
20 mkdir -p test
21 cd test
22 mpirun ../$task                # generate input file input.json
23                               # run the simulation
24
25 python3 $pp                    # script in the directory of the output
```

1-9: #SBATCH --option define resources selection (e.g. partition. number of cpus and memory).

10-13: setup the variables to use mpi (parallel codes) and gcc compiler libraries (c/c++).

15-16: set variables with the name of exercise folder you want do (e.g. `attractive_potential`,

froehlich) and the python script.

18-24: enter in the exercise folder, create a folder named `test` (if present nothing is done) and enter it.

Generate the input file `input.json` with the first call to `mpirun ../$task`.

Run the code with `mpirun ../$task`, produce the output in the current folder and analyze it with python.

- 7. SSH with graphic support:** The following `ssh` command adds options to avoid disconnections from the server and enable the graphic server to display results.

```
ssh -o ServerAliveInterval=60 -X -J name.surname00@bastion-nav.difa.unibo.it name.surname00@ophfe1
```

The command is prompted in a single line. Breakline in the box is only for visualization.

- 8. Display images on the cluster:** After enabling graphic support (section 7) you can display images located on the cluster directly on your screen.

```
xdg-open my_image.pdf
```

This command will open a graphical editor available on the cluster and show you the image.

This guide shows the necessary steps to do a Diagrammatic Monte Carlo calculation in a cluster.

Further informations and details on how to modify the jobs, automatize the calculations will be provided during the lab.